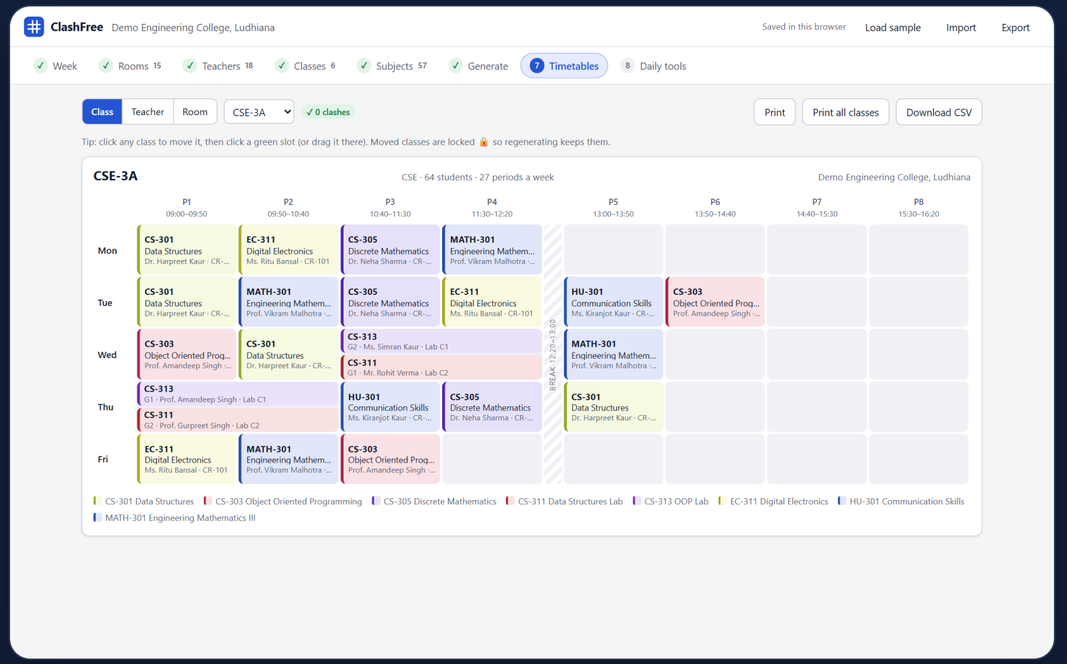
HACKATHON 1 · SCHEDULING OPTIMIZER

ClashFree Timetables

Clash-free college timetables in seconds, for any college.

Husanpreet Singh · Punjab Jobs AI Bootcamp · J-PAL South Asia

Live: hussnn18.github.io/h1 · Code: github.com/Hussnn18/h1



THE PROBLEM

Timetables are still built by hand, and they still clash

Weeks of work

Every semester each department fits its sections, labs and teachers into the week by hand.

Clashes and uneven loads

Double-booked teachers and rooms; heavy days for some teachers, empty days for others.

Shared rooms fought over

Departments compete for seminar halls and computer labs, while other rooms sit empty.

Who it is for: the timetable in-charge of any college. Goal: a clash-free timetable without help or training.

HOW IT IS BUILT

One web page that does everything in the browser

1 Enter
Week, rooms, teachers, classes, subjects. Or paste from Excel.

2 Save
One JSON file kept in the browser. Export and import it.

3 Solve
Check data, place hardest first, repair, retry. Off the main thread.

4 Verify
A separate checker re-tests every rule before showing “0 clashes”.

5 Use
Grids, moving by hand, print, CSV, cover for absent teachers.

No server

Plain HTML, CSS and JavaScript modules (about 2,150 lines of JS). Hosted free on GitHub Pages.

16 automated tests

Every sample solves with 0 clashes; every kind of clash is caught by the checker.

Built with Claude Code

Planned, written, tested and debugged with Claude Code, in small Git commits.

WHAT IT LOOKS LIKE

Eight steps, one timetable

1 Step bar

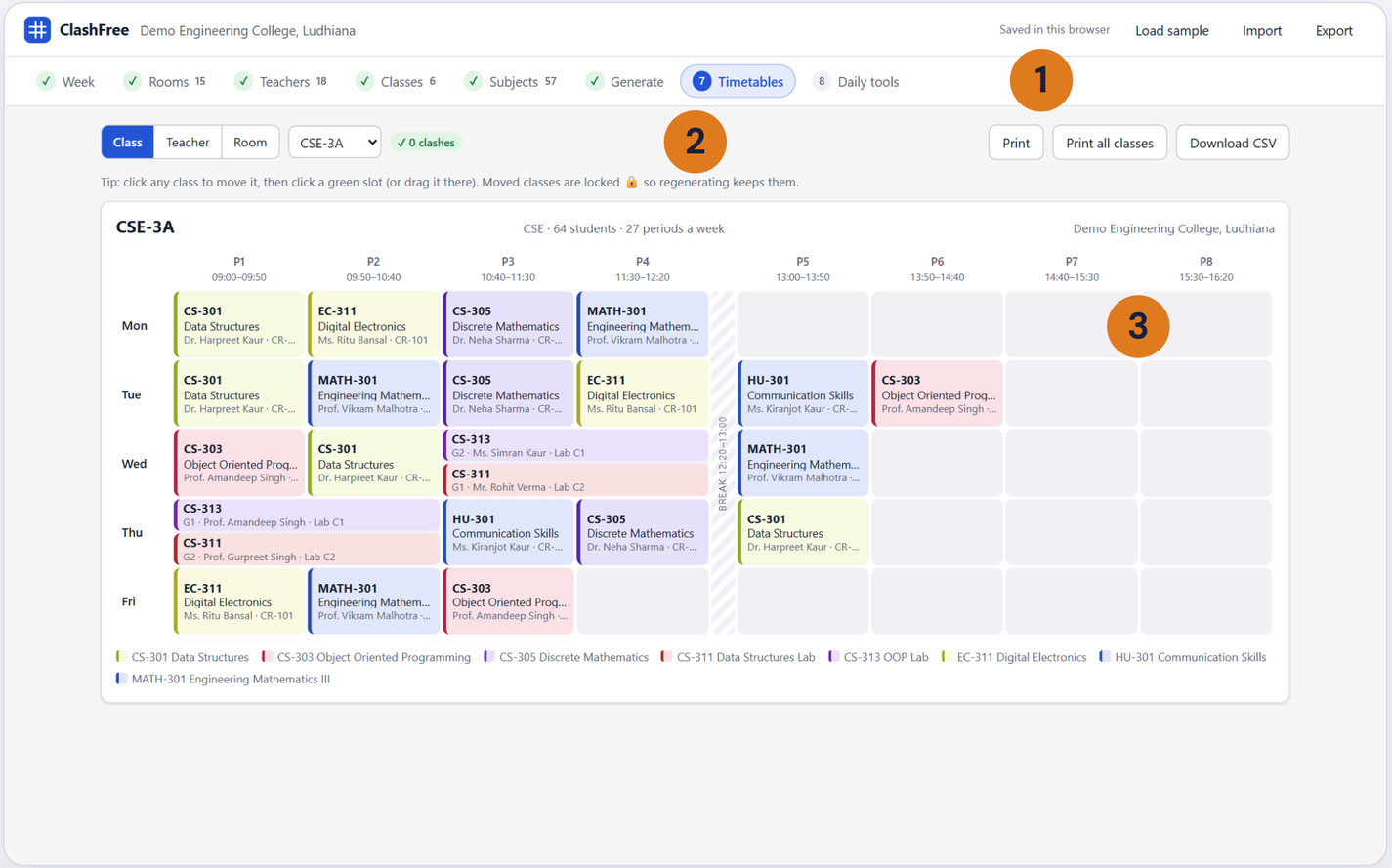
Week → Rooms → Teachers → Classes → Subjects → Generate → Timetables → Daily tools. A ✓ appears when a step is done.

2 Toolbar

View by class, teacher or room. The green badge is the clash count from the checker. Print and CSV.

3 Timetable grid

Days × periods with times and the lunch break. One colour per subject; labs span two periods.



HOW TO TEST IT

Test it in five minutes, no sign-up

LIVE APP

hussnn18.github.io/h1

CODE (PUBLIC)

github.com/Hussnn18/h1

Steps for evaluators

1. Open the app; choose “Engineering college”.
2. Generate → Generate timetable: 133 lessons, 0 clashes.
3. Open timetables; switch Class / Teacher / Room.
4. Click a class, then a green slot, to move it.
5. Daily tools: pick a teacher and day for cover.

Test datasets (all fictional)

Load sample →	What it shows
Engineering college	4 departments, shared labs, split batches
Small college	Learn the app in two minutes
Large institute	20 classes, 440 lessons, about 3 s
Data with mistakes	The plain-language error report

Same files in the repo: samples/*.json and samples/csv/.

FEATURE 1 · STEP 1: WEEK

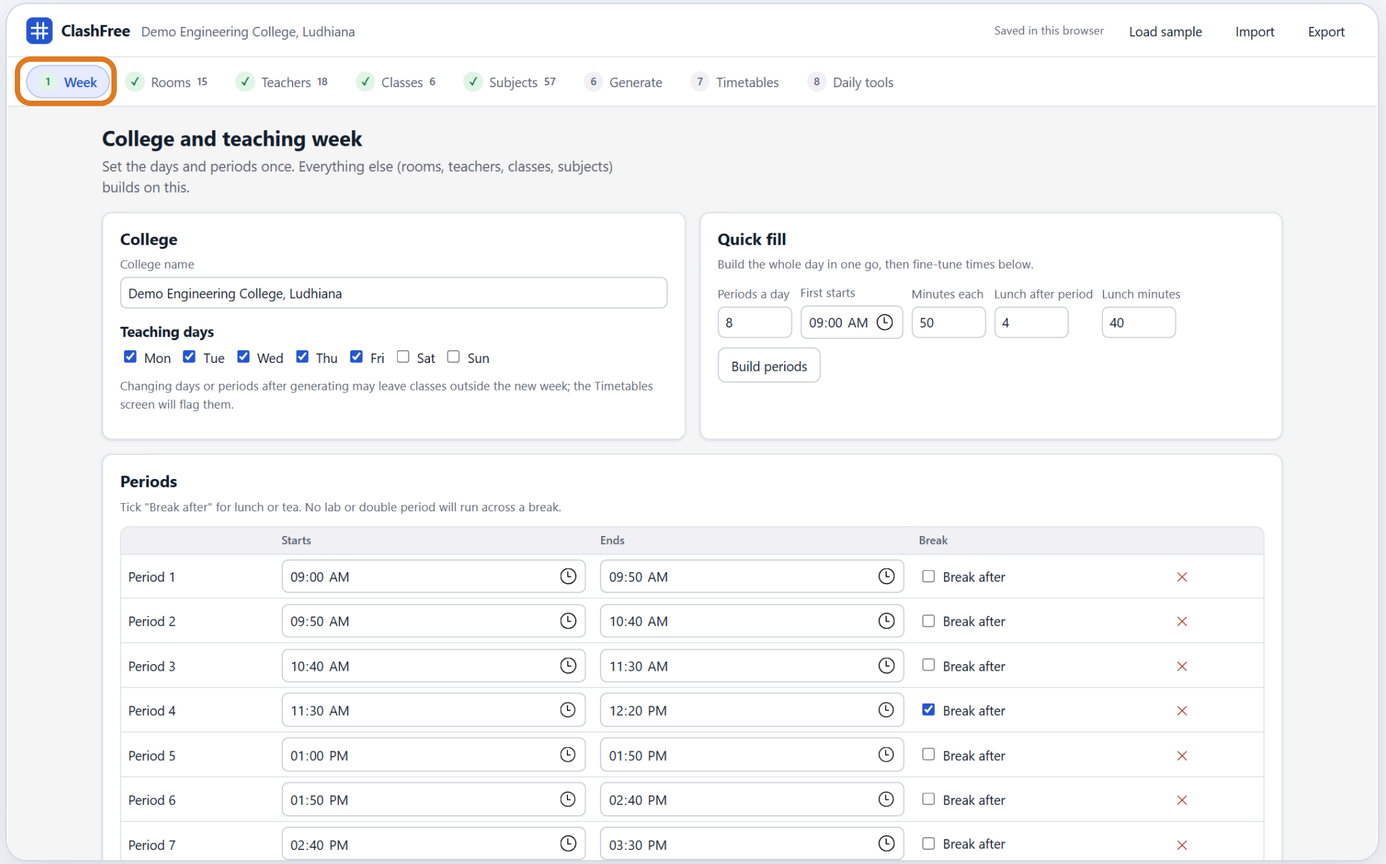
Set the teaching week once

How to use

1. Open Week in the step bar.
2. Tick the teaching days.
3. Quick fill: periods, start time, length, lunch → Build periods.
4. Adjust times; tick “Break after” for lunch or tea.

How it is designed

Any college: 5 or 6 days, any periods. No lab or double period may run across a break.



FEATURE 2 · STEPS 2-3: ROOMS, TEACHERS

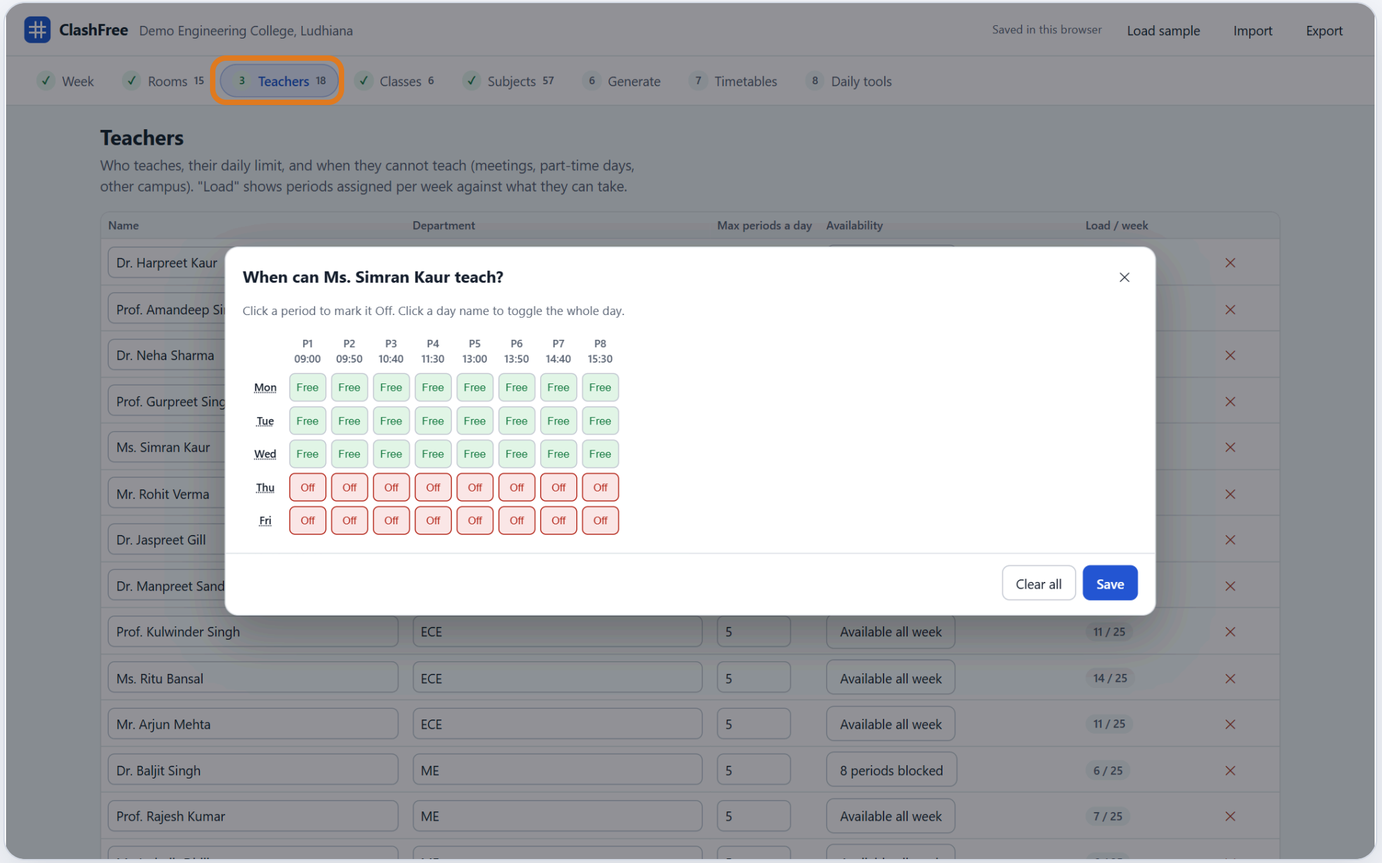
Rooms, teachers and when they are free

How to use

1. Rooms: name, type, seats, department (blank = shared).
2. Teachers: department, max periods a day.
3. Click “Available all week” to mark periods Off.
4. Load / week turns red if a teacher is overbooked.

How it is designed

Department rooms serve only their department; shared rooms (seminar hall, computer centre) serve all.



FEATURE 3 · STEPS 4-5: CLASSES, SUBJECTS

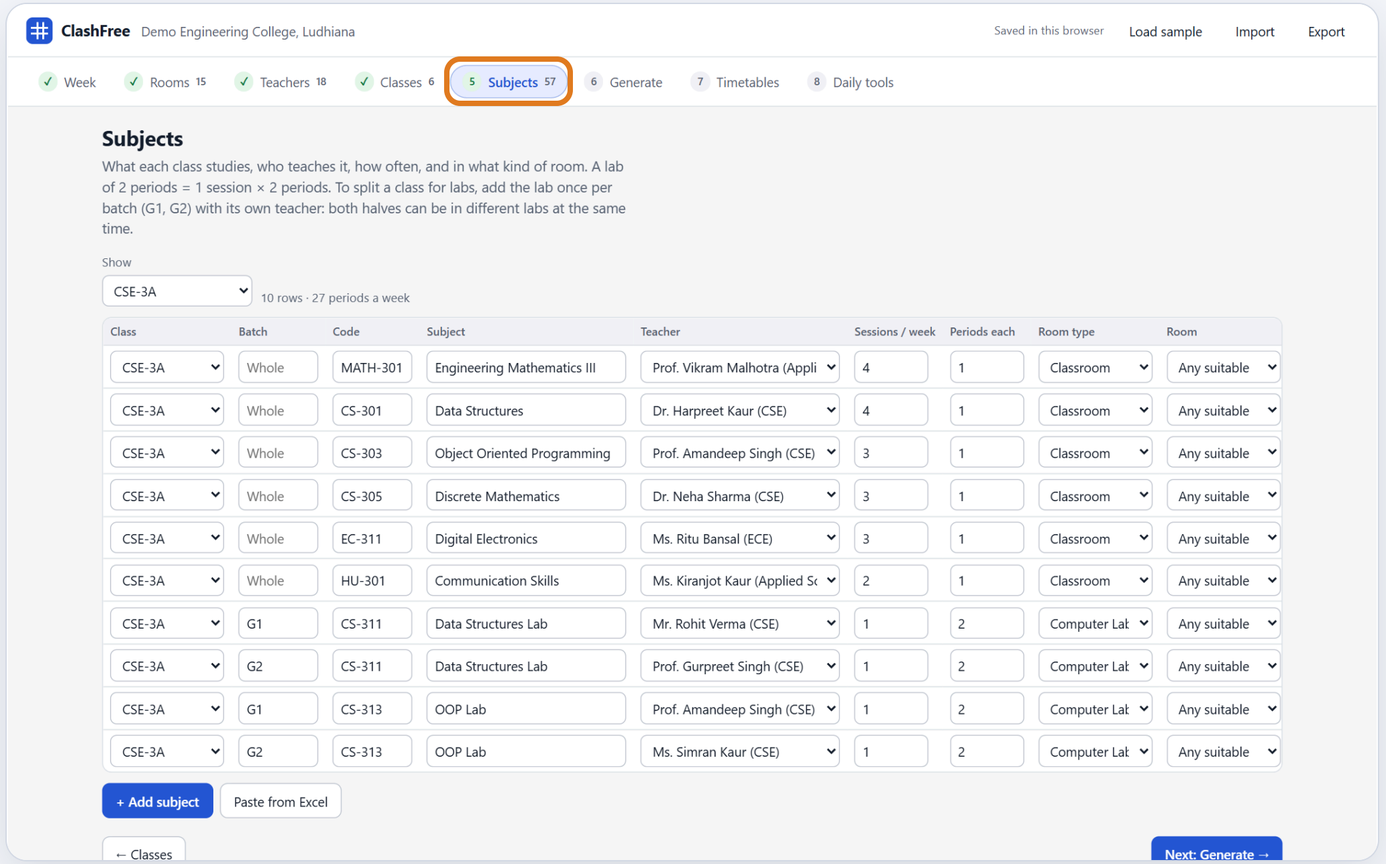
Classes, subjects and split lab batches

How to use

1. Classes: name, department, students.
2. Subjects: class, teacher, sessions a week, room type.
3. A 2-period lab = 1 session × 2 periods each.
4. Split a class: add the lab once per batch (G1, G2).

How it is designed

G1 and G2 may be in two labs at once; the solver lines them up so neither half sits idle.



FEATURE 4 · UNDER EVERY LIST

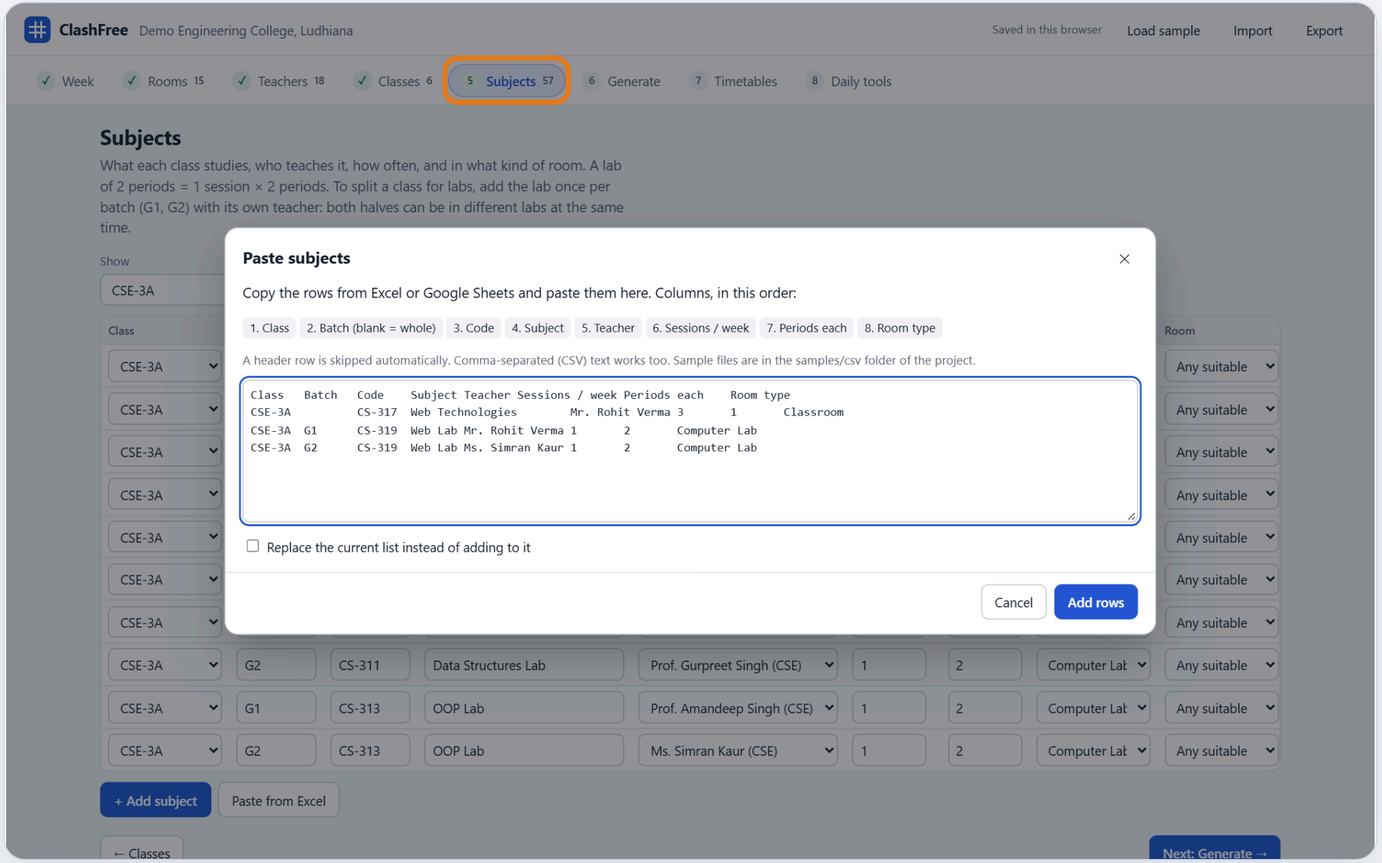
Paste lists straight from Excel

How to use

1. Click “Paste from Excel” under any list.
2. Copy rows in Excel or Google Sheets; paste.
3. The header row is skipped; CSV works too.
4. Click “Add rows”, or tick Replace first.

How it is designed

Subjects name their class and teacher; missing ones are created, so nothing is lost.



FEATURE 5 · STEP 6: GENERATE

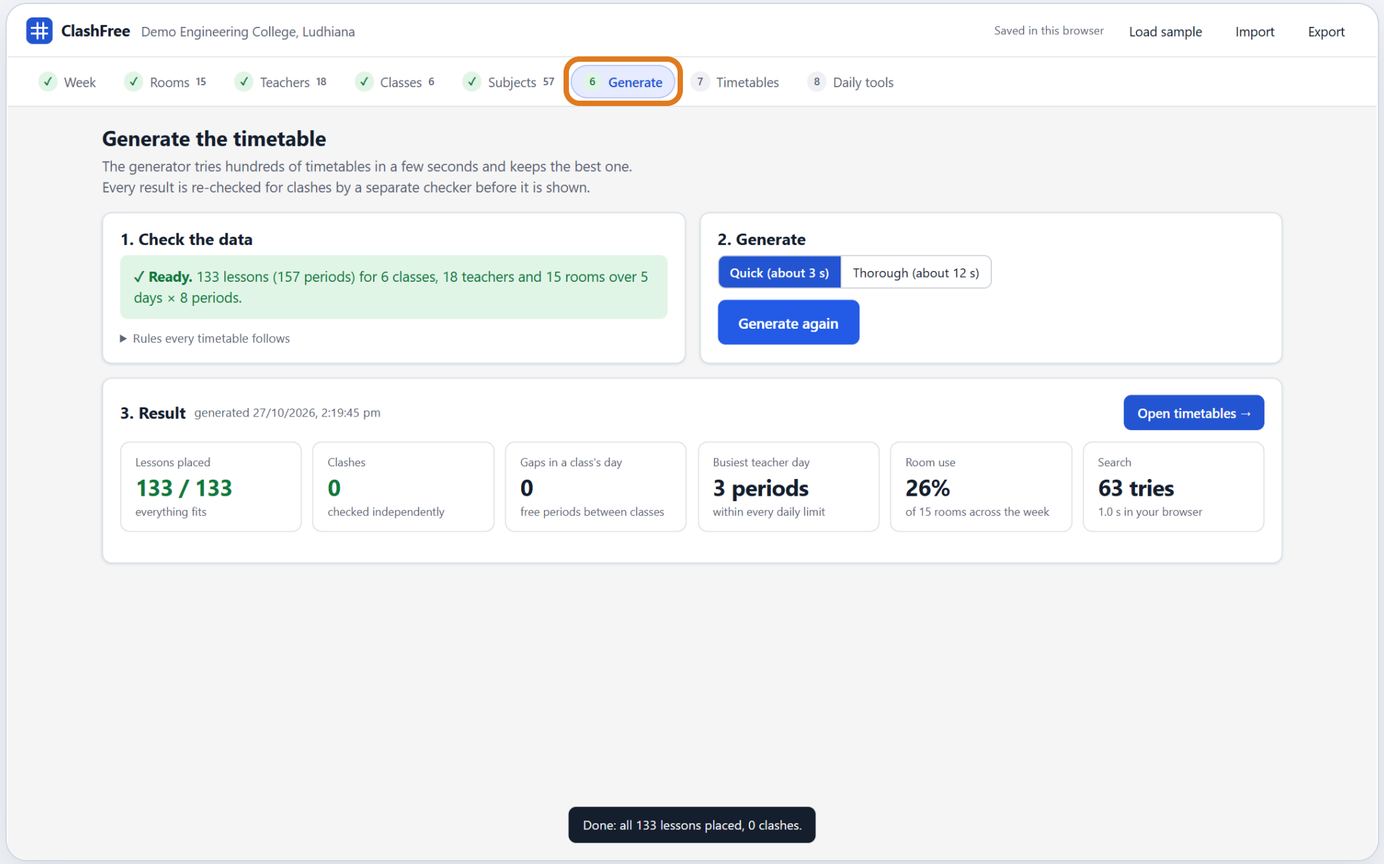
Generate in a few seconds

How to use

1. Open Generate; a green box means the data is ready.
2. Choose Quick (about 3 s) or Thorough (12 s).
3. Click “Generate timetable”.
4. Read the result cards, then Open timetables.

How it is designed

Hardest lessons first, each to its best legal slot; stuck ones move 1–2 others. Hundreds of tries; best kept.



FEATURE 6 · STEP 6: GENERATE

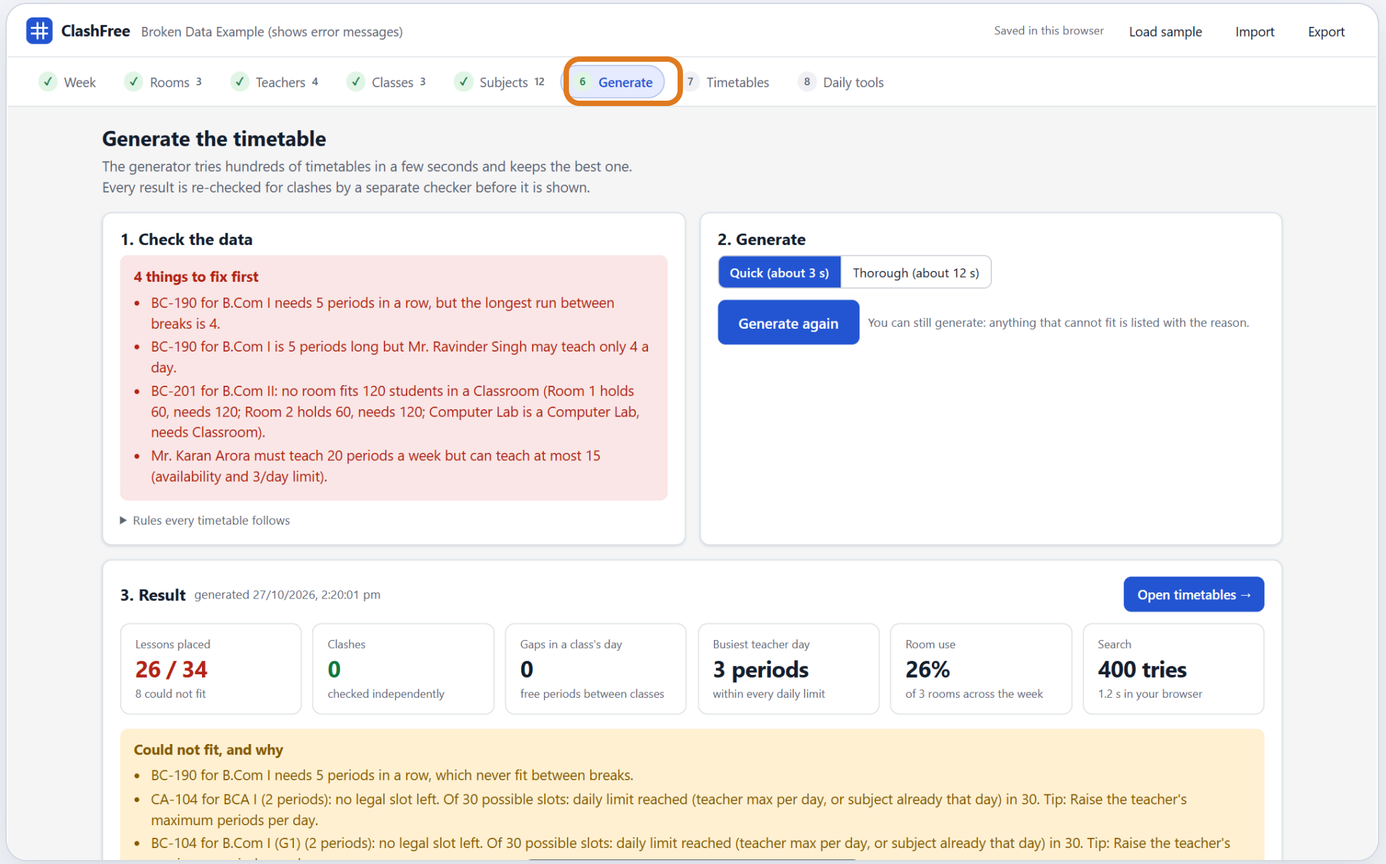
Plain-language errors, never silent gaps

How to use

1. Load sample → “Data with mistakes”.
2. The red box lists what to fix, in plain words.
3. Generate anyway: everything that fits is placed.
4. Each class that did not fit shows why, plus a tip.

How it is designed

A check before solving catches overloaded teachers, classes bigger than every room and labs longer than a block.



FEATURE 7 · STEP 7: TIMETABLES

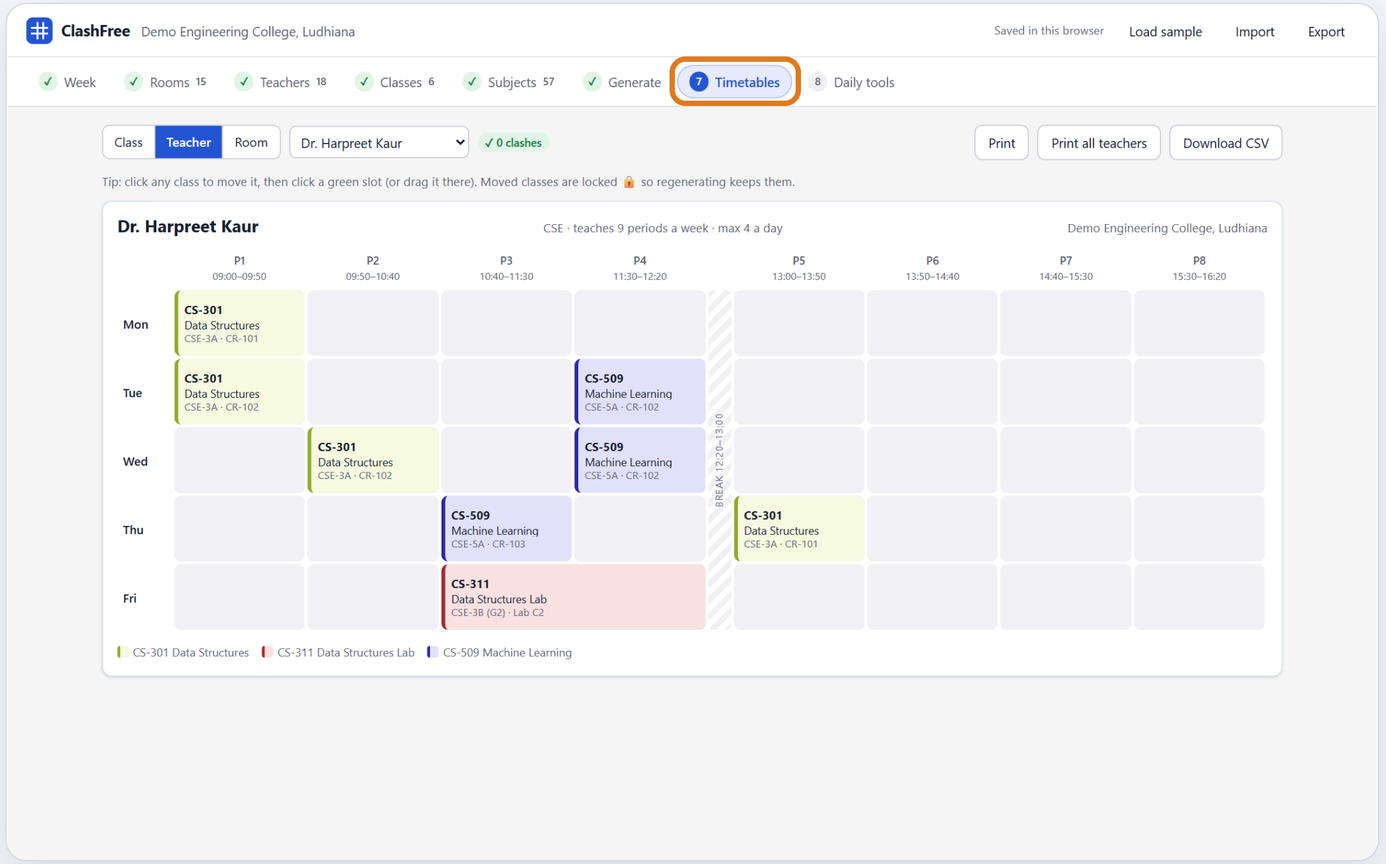
Every class, teacher and room

How to use

1. Open Timetables.
2. Choose Class, Teacher or Room.
3. Pick one from the list, grouped by department.
4. Hover any class for its full details.

How it is designed

One timetable, three views. Labs span their periods; parallel batches sit stacked in one cell.



FEATURE 8 · STEP 7: TIMETABLES

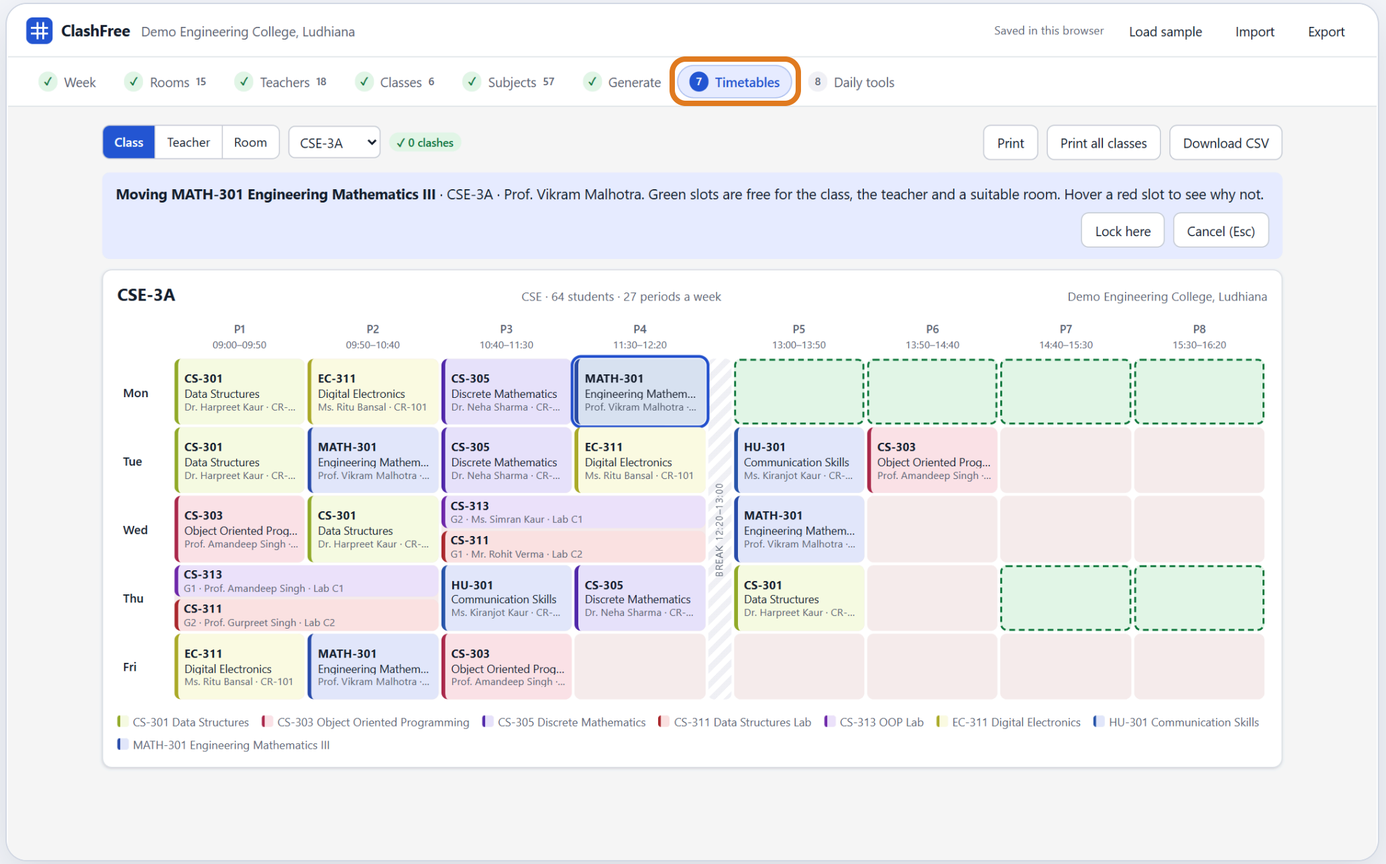
Move a class by hand, safely

How to use

1. Click a class in the grid (or drag it).
2. Green slots are free; hover a red one to see why not.
3. Click a green slot. Esc cancels.
4. The class is now locked: Generate again keeps it.

How it is designed

Every slot is tested live against the solver's rules: class, teacher, room, availability, daily limits.



FEATURE 9 · TIMETABLES TOOLBAR

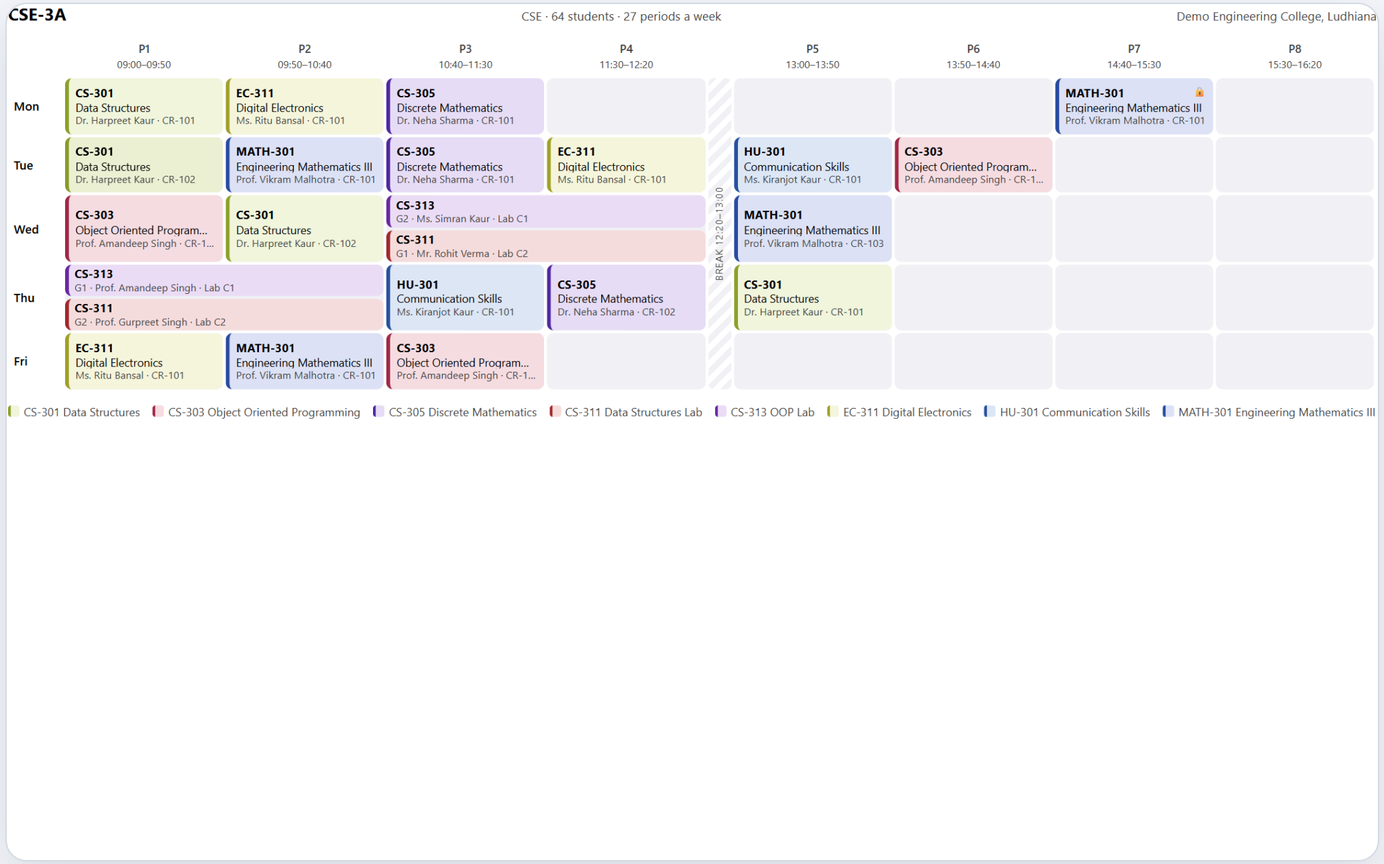
Print, PDF and Excel

How to use

1. Print: the timetable on screen.
2. Print all: one per A4 page; pick Save as PDF.
3. Download CSV: every class in one Excel sheet.
4. Export / Import (top right): a backup file.

How it is designed

The print layout hides every button, fits A4 landscape and keeps the subject colours.



FEATURE 10 · STEP 8: DAILY TOOLS

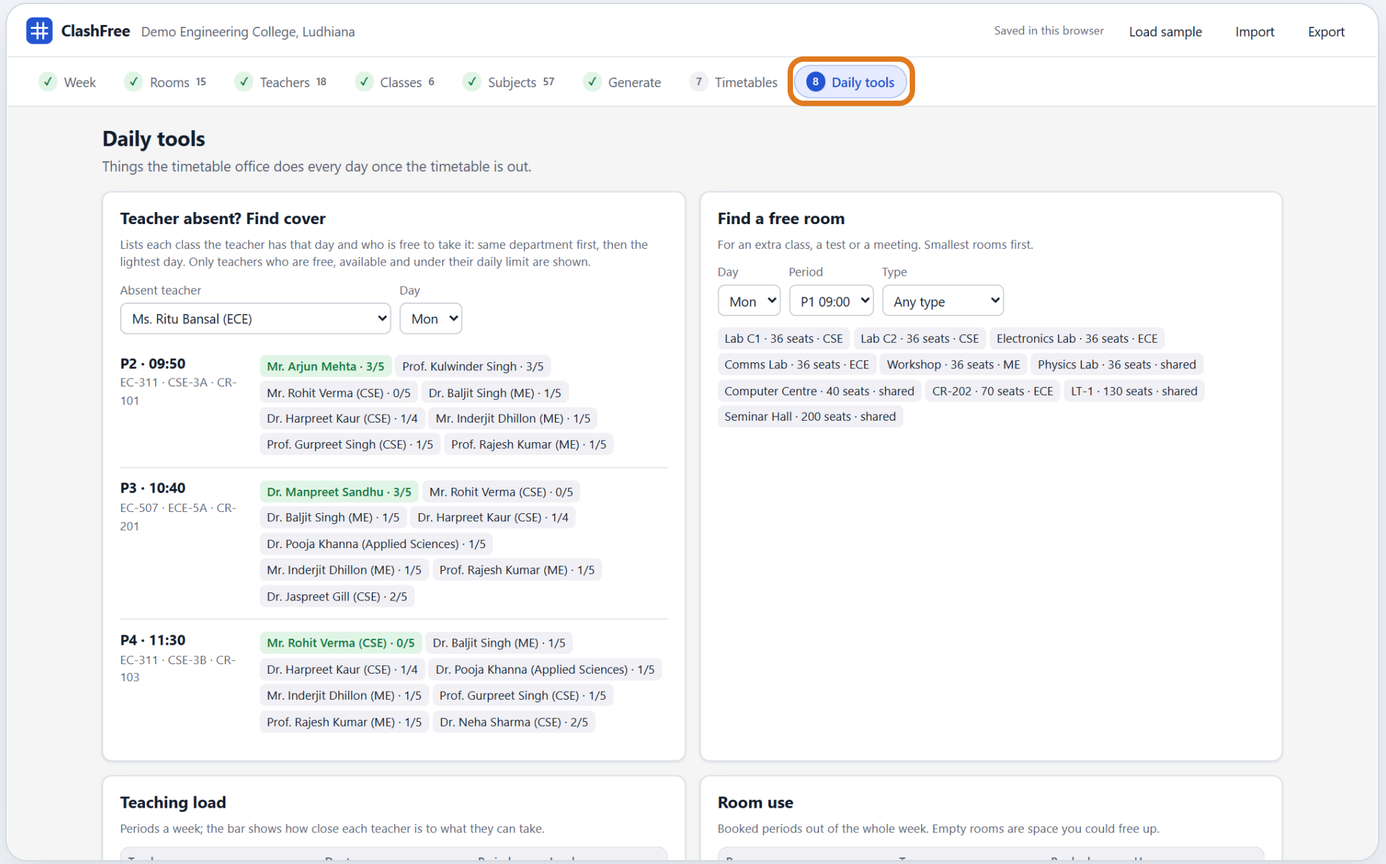
Cover absent teachers, find free rooms

How to use

1. Pick the absent teacher and the day.
2. Each of their classes lists who can cover; green = best.
3. Find a free room by day, period and type.
4. Teaching load and room use are below.

How it is designed

Cover must be free, available and under the daily limit; same department first, then the lightest day.



WHAT MAKES A TIMETABLE GOOD

Rules it never breaks, and what it prefers

Never broken

- No teacher, room or class in two places at once
- Teachers only when available, within their daily maximum
- Right room type, enough seats, own department or shared
- Labs stay together, never across a break
- A subject at most once a day (unless sessions exceed days)

Preferred, and scored

- No free gaps inside a class’s day
- An even load across the week, for classes and teachers
- Split lab batches side by side
- Fewer last-period classes; the smallest room that fits
- Classes locked by hand stay where they are

RESULTS

Tested at college scale

0 clashes

in every sample, confirmed by the separate checker

133 / 133

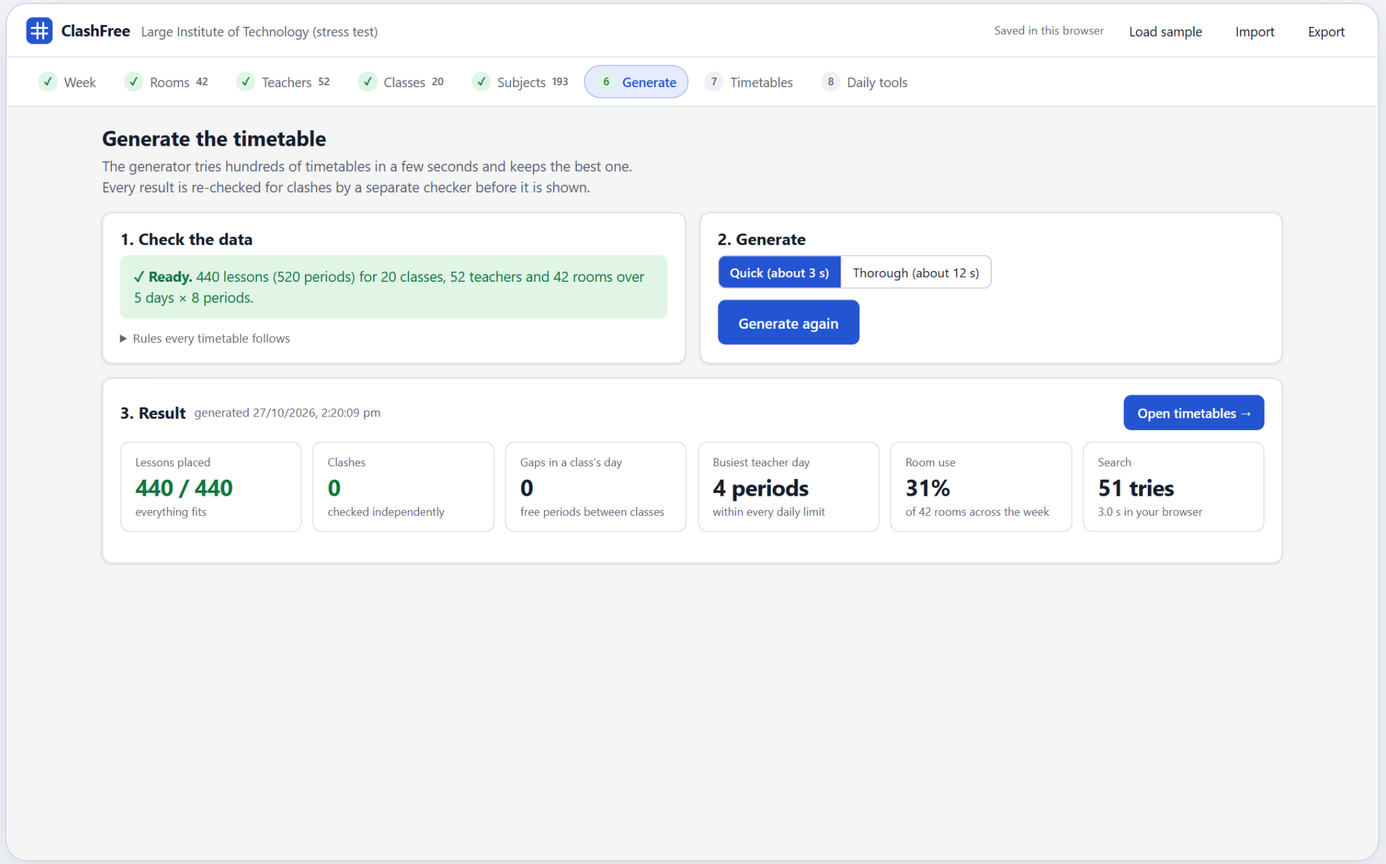
lessons placed for the engineering college demo

440 lessons

20 classes, 52 teachers, 42 rooms, placed in about 3 s

16 / 16

automated tests pass; works on a phone too



TRY IT

A clash-free week, in one click

LIVE APP

hussnn18.github.io/h1

CODE, README, CLAUDE.md, DATASETS

github.com/Hussnn18/h1

What comes next

- Combined classes: one elective for several sections
- Teacher preferences, such as no first period
- One shared online copy for several admins (today: one browser, plus Export / Import)